Information or Accountability?

The 1869 Repeal of the Dutch Newspaper Stamp and Parliamentary Speech

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Representatives speak for those who can vote. This paper asks whether a cheaper press let them speak for people who could not. It answers with a tax shock to the price of news that left the franchise rules unchanged, in a parliament with one chamber that faced voters and one that did not.

Question

When the franchise widened, public spending moved towards the new voters, in American states that dropped poll taxes and literacy tests or gave women the vote (Husted and Kenny 1997; Miller 2008). People without the vote could still be heard through the press. Parliamentary attention tracks the attention of newspapers (Walgrave et al. 2008; Vliegenthart et al. 2016). Newspapers cover what sells to readers and advertisers (Strömberg 2004), and a reader need not be a voter.

Whether a cheaper press gives a voice to those without the vote depends on why parliaments follow it. If newspapers are a cheap source of information about the country, whether readers can vote does not matter. If they reach the audience to which legislators are accountable, it does. Elites describe the press as both an information source and an arena (Van Aelst and Walgrave 2016), and today most readers vote, so the two channels are observationally equivalent. The evidence that parliaments follow the press starts after 1945 (Tan and Weaver 2007), and for the Netherlands the one study of the years before 1990 finds no media-led agenda-setting (Kleinnijenhuis and Rietberg 1995). The economics of media and accountability has the identification strategies this question needs: coverage raises legislator effort and newspaper entry raises turnout (Snyder and Strömberg 2010; Gentzkow et al. 2011). The electoral channel has been isolated for judges, whose sentences respond to coverage only where judges are elected (Lim et al. 2015), and within elected chambers through proximity to elections (Arceneaux et al. 2016). No study of legislators has used a chamber that reads the same press and does not face voters.

Setting

On 1 July 1869 the Dutch state abolished the *dagbladzegel*, the stamp duty on every copy of a newspaper, together with the duty of *f* 0.35 on every advertisement (law of 9 April 1869). The stamp had taken up to 57 per cent of the subscription price and on average a quarter of production costs. In 1851 the largest title, the *Algemeen Handelsblad*, printed 5,400 copies, while most papers appeared one to three times a week for a few hundred subscribers (Aerts 2019). After July 1869 publishers enlarged their papers, went daily and sometimes cut prices (Hemels 1969, p. 268), and the press kept growing for decades (van Kranenburg et al. 1998). In the booksellers' address books, which I have digitised from Hemels (1969), the number of titles rose from 163 in 1869 to 203 in 1870 and 553 in 1890, and the titles appearing six or seven times a week rose from 15 to 29 in one year. The twelve titles digitised throughout appeared about 30 per cent more often (Figure 1).

The franchise did not move with the price of news. The electoral law of 1850 tied the vote to a

municipal tax census fixed by statute until the revision of 1887, first applied at the election of 6 March 1888 (de Jong 2017). Registered voters grew only from about 91,000 in 1860 to 134,000 in 1887, and until 1888 only one adult man in eight or nine could vote. Reading was far more common. Around 1869 seven bridegrooms in eight and three brides in four could sign their marriage certificate, and more could read (Boonstra 1993, pp. 21–22, 104; Vandezande and Matthijs 2009).

The two chambers read the same newspapers and debated the same bills. The Tweede Kamer was elected directly under the census, half of it renewed every two years. The Eerste Kamer was chosen by the Provincial States from the most heavily assessed taxpayers for nine-year terms, so its members did not face voters.

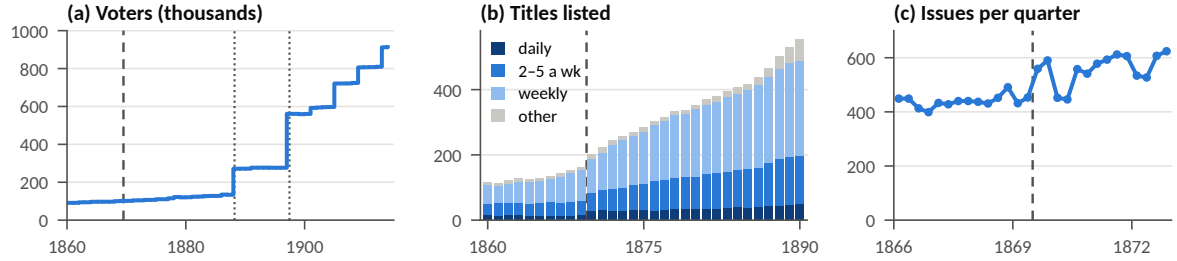


Figure 1. Voters and newspapers. (a) Registered voters, summed over electoral districts. (b) Newspaper titles in the booksellers' address books by frequency of appearance. (c) Issues per quarter of the twelve titles digitised in every year 1867–1871. Dashed lines: 1 July 1869. Dotted lines: first elections after the franchise extensions of 1887 and 1896. Sources: Huygens ING, Hemels (1969) digitised for this project, and Delpher.

Data

- **Parliament.** The proceedings (*Handelingen*) of both chambers from Staten-Generaal Digitaal, harvested for 1848–1940 (214,254 OCR pages) and segmented into 336,747 speeches. Speeches are linked to members by chamber, sitting date and service spells from Wikidata, PoliticalMashup and the person identifiers of the election returns. Party families are coded by hand. A blind-coded stratified sample of pages measures attribution accuracy by decade and chamber.
- **Press.** The article-level open archive of the Koninklijke Bibliotheek (Delpher), 1848–1879: 4.8 million articles with title, date, place, distribution area and type. Nine titles that went on appearing after 1869 are missing from the archive from 1870, so press measures use the twelve titles digitised in every year 1867–1871.
- **Titles.** Every title in the booksellers' address books, 1851–1890, with publisher, place, days of appearance and price, digitised from Hemels (1969) as a title-year panel of 1,237 titles.
- **Districts.** Election returns with candidate votes (Huygens ING, Kiesraad), municipal data (HDNG, censuses) and boundaries (NLGIS).

Measurement

The main measure of press-following needs no topics. As in Gentzkow and Shapiro (2010) and Gentzkow et al. (2019), speech and press are represented by bigrams of normalised words, on a vocabulary fixed in advance: the phrases that both the press and parliament used before 1869. For member i in month m, with phrase frequencies q_{im} and press frequencies P_m,

$$s_{im} = \cos(q_{im}, P_{m-1}) - \tfrac{1}{|K|} \textstyle\sum_{k \in K} \cos(q_{im}, P_{m+k}),$$

where K holds the months within a year of m other than $m-1$, m and $m+1$, so stable style, slow drift in spelling and OCR, and ever-present topics cancel out. The press vectors leave out reports of parliamentary proceedings, which reproduce speeches.

A topic index asks whether members take up the policy areas the press covered: $f_s = \sum_a p_{sa}(A_{a,m-1} - \bar{A}_a)/\mathrm{sd}(A_a)$, with $A_{a,m}$ the press share of area a. Policy areas follow the Comparative Agendas Project codebook (Bevan 2019) with six nineteenth-century codes. A large language model coded 39,496 speeches and articles without seeing date, chamber or speaker, so the coding rule cannot change at the reform, and two human coders code a stratified subsample blind. Design-based supervised learning corrects the estimates for the model's errors (Egami et al. 2023). The labels train a classifier fine-tuned from GysBERT (Manjavacas Arevalo and Fonteyn 2022) after continued pre-training on in-domain text (Debaene et al. 2025), and topic probabilities carry classification uncertainty into the estimates.

Empirical strategy

The core design is a bicameral event study. For member i in chamber c and half-year t,

$$y_{ict} = \alpha_{ic} + \lambda_t + \sum_{h \neq -1} \beta_h \, \mathbf{1}[c = \mathrm{TK}] \, \mathbf{1}[t - t_{1869} = h] + \varepsilon_{ict},$$

on the two years either side of the reform (1867H2–1871H1), with one and three years and the whole period 1850–1879 as robustness. Member-by-chamber effects absorb composition, half-year effects absorb the whole media environment, and β_h traces the lower chamber's differential response, with the pre-reform coefficients as a test of parallel trends. Because treatment varies across two chambers only, inference rests on the chamber-difference series, with Newey–West errors and placebo-date randomisation over every half-year whose window does not overlap the reform, a placebo-in-time analogue of Conley and Taber (2011) and Hagemann (2019). Estimates are reported for all members and for members who served on both sides of July 1869, which guards against the bias of two-way fixed effects under heterogeneous effects (de Chaisemartin and D'Haultfœuille 2020; Sun and Abraham 2021).

Three further contrasts sharpen the mechanism. Members whose seat was up in June 1871 are compared with those elected in June 1869. Members from districts where the press grew after 1869 are compared with the others, with entry instrumented by pre-1869 market size $\times$ Post, pre-1869 advertising intensity held fixed, and wild-cluster bootstrap inference over 38–43 districts (Cameron et al. 2008). The response should rise with the reading public outside the electorate, measured as school attendance in 1870 (Knippenberg 1986) relative to registered voters. Press-following towards the titles census voters read is compared with the cheap titles, with the stamp's share of each title's pre-1869 price (Hemels 1969) as the dose. Lead–lag local projections (Jordà 2005) show which series moves first, by chamber and regime.

Threats

- **The periodic election of 8 June 1869**, three weeks before the reform, renewed half the lower chamber in the year the table of seats was revised. A cycle-matched placebo uses only the half-years that follow a periodic election.
- **The advertisement duty** fell with the stamp. The district design holds pre-1869 advertising intensity fixed.
- **OCR quality** may differ between the two chambers' printed proceedings. An OCR index enters as a control and as a sample restriction, and a debate-level version free of speaker attribution is estimated alongside.

- **The constitutional conflict of 1866–68**, which made cabinets answer to the lower chamber, falls in the pre-period. The pre-reform coefficients and a version that drops those years test it.
- **The Franco-Prussian war** filled the press of late 1870. Press-following is also measured without international affairs and without late 1870.

Hypotheses

- **H1, information or accountability.** If legislators follow the press because its readers can sanction them, the shock raises press-following more in the Tweede Kamer than in the Eerste Kamer, more among members facing re-election soon, and more towards the titles voters read. If they follow it as a source of information, the response does not vary with electoral exposure.
- **H2, whose issues.** If the cheaper press gave a voice to readers without the vote, the shock raises attention to issues with large non-voting constituencies (labour, poor relief, public health, housing, the franchise), most in response to the cheap titles. These issues reached voters through what they paid: under the Poor Law of 1854 municipal funds relieved the poor where church charity fell short, and relief was a strategy of the propertied as well as of the poor (van Leeuwen 2000). A response confined to the lower chamber means that this voice worked through voters.

Pilot evidence and status

The specifications, outcomes and inference go into a pre-analysis plan to be registered in November 2026, which lists every pilot estimate and precedes the validated classifier and the human-coded sample. Pilot estimates with a baseline classifier have the sign H1 predicts. After July 1869 press-following rises in the lower chamber relative to the upper by 0.42 standard deviations of press attention (Figure 2a), by 0.45 with every policy area and by 0.48 with the titles that entered the archive after the reform. In both chambers attention to a policy area rises in the quarter after the press's, by more and for longer than the press's rises after parliament's (Figure 2b). Testing H2 needs the newspapers missing from the open archive after 1869.

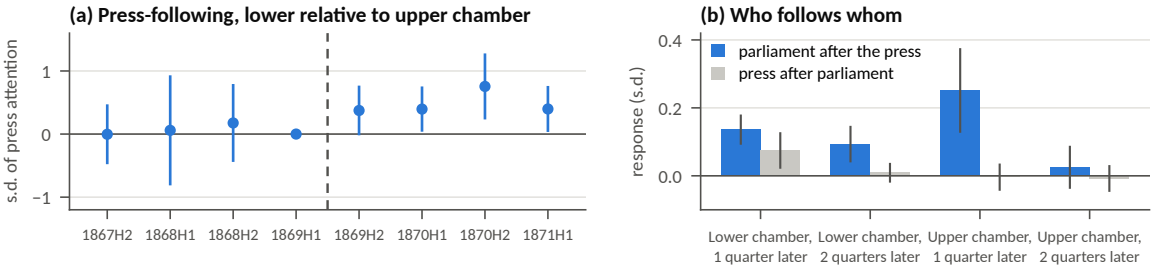


Figure 2. Pilot estimates, baseline classifier. (a) The event study by half-year against 1869H1, 95 per cent intervals clustered by member. The index uses domestic policy areas (the war of 1870 filled the foreign news). Dashed line: 1 July 1869. (b) Local projections, 1850–1879: response to a rise of one standard deviation in the other's attention to a policy area, with 95 per cent Driscoll–Kraay intervals.

Contribution

The paper compares a chamber that faces voters with one that does not while holding the press fixed, a design so far used for judges. To my knowledge it is the first causal study of the repeal of a newspaper tax. The repeals in Belgium (1848), Britain (1855) and the Netherlands

are documented in descriptive press histories (Luykx 1955; Hewitt 2014; Hemels 1969), and the closest causal evidence concerns other shocks to the cost of news (Perlman and Sprick Schuster 2016). If the channel is electoral, much of what the agenda-setting literature attributes to media influence is a statement about accountability, and a cheaper press widens representation only where its readers can vote.

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